

Notes: Asriyan and Vanasco (RES, 2024)

Security Design in Non-Exclusive Markets with Asymmetric Information

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1 Big picture

- **Question.** How does optimal security design change when an informed seller cannot commit to trade exclusively with a single buyer?
- **Setting.** A risk-neutral seller privately knows whether her asset is high or low quality. Many competitive, uninformed buyers post menus of security–price contracts. The seller can accept contracts from multiple buyers subject to the asset’s cash-flow capacity.
- **Core challenge.** In the standard exclusive-contracting literature, the informed seller can signal quality by retaining part of the cash flow. Under non-exclusivity, however, that retention may be undone by another buyer who offers to buy the “retained” part.
- **Main conceptual contribution.** The paper shows that the classic separation logic from exclusive security-design models breaks down once markets are non-exclusive. Non-exclusivity changes the relevant incentive-compatibility constraint and forces semi-pooling.
- **Main benchmark result.** With full information, non-exclusivity is irrelevant: all asset cash flows are sold and priced at full-information expected value.
- **Main exclusive-markets result.** If the seller can accept at most one menu, equilibrium is unique. When the prior probability of high quality is low, equilibrium is separating: the high type issues debt and retains the tail, while the low type sells the full claim. When the prior is high, equilibrium still uses debt for the high type but includes cross-subsidization to the low type.
- **Main non-exclusive-markets result.** Separating contracts are never part of equilibrium. There is always a semi-pooling equilibrium—the *star-equilibrium*—in which both seller types issue the same senior debt tranche at average valuation, while only the low type sells the junior tranche at low valuation.
- **Why this matters.** The seller faces more mispricing under non-exclusivity than under exclusivity. Market liquidity can go either way, but informational rents and the structure of tranching change sharply.
- **Main welfare result with exogenous asset quality.** Non-exclusive markets can be better or worse than exclusive markets. They help when they raise liquidity enough, and hurt when they reduce liquidity.
- **Main welfare result with endogenous asset quality.** Once asset quality depends on costly origination effort, non-exclusive markets always produce *lower* average asset quality than exclusive markets because they compress the payoff gap between high- and low-quality assets.
- **Empirical relevance.** The model rationalizes tranching, the sale of retained pieces, and the weak link between retention and senior-tranche pricing seen in mortgage-backed securities markets, especially where exclusivity is hard to enforce.
- **Economic takeaway.** Opacity and non-exclusivity can increase trade in some states of the world, but they also increase mispricing and weaken incentives to originate high-quality assets.

2 Data and measurement (key objects)

2.1 Why this is a theory paper but still has “key objects”

- This is not a reduced-form or structural estimation paper. There is no sample construction, no regression panel, and no instrument.
- The paper’s “data” are the model objects that determine equilibrium: seller type, cash-flow distributions, securities, menus, beliefs, and payoffs.
- The key measurement task is therefore conceptual: define the tradable claims, define the constraints that keep those claims feasible, and define the equilibrium objects that can be compared across exclusive and non-exclusive market structures.

Interpretation. The note keeps the same study-note structure as an empirical paper, but here the “measurement” section means: what exactly is the model economy, and what are the objects whose movement the propositions explain?

2.2 Agents, timing, and preferences

There are two dates, $t \in \{1, 2\}$, one seller, and a large number of competitive buyers.

Seller utility:

$$U^S = c_1^S + \delta c_2^S, \quad \delta \in (0, 1).$$

Buyer utility:

$$U^B = c_1^B + c_2^B.$$

- The seller is more impatient than the buyers, so gains from trade arise even absent risk sharing.
- The seller owns an asset that pays random cash flow X at date 2.
- The asset quality is $\theta \in \{H, L\}$ and is privately known by the seller.
- Buyers only know the prior belief $\mu_0 = \Pr(\theta = H)$.

Interpretation. The motive for trade is simple discount-factor heterogeneity; all of the interesting frictions come from asymmetric information and non-exclusivity.

2.3 Asset quality and the information structure

- The asset cash flow under type θ has distribution G^θ with density g_θ on $[0, \bar{X}]$.
- The model imposes monotone likelihood ratio ordering:

$$\frac{g_H(x)}{g_L(x)} \text{ is increasing in } x.$$

- Higher cash flows are therefore more informative of high quality.

Interpretation. This assumption gives the usual ranking needed in security-design models: high types value payoff profiles tilted toward high cash flows more than low types do.

2.4 Securities and feasibility constraints

A security is a function

$$F : [0, \bar{X}] \rightarrow \mathbb{R},$$

where $F(X)$ is the payment promised to the buyer when asset cash flow X is realized.

If the seller issues a collection of securities $\mathcal{F}$, feasibility requires:

- **Capacity constraint (CC):**

$$\sum_{F \in \mathcal{F}} F(x) \leq x, \quad F(x) \geq 0 \text{ for all } x.$$

- **Weak monotonicity (WM):**

$$F(x) \text{ and } x - \sum_{F \in \mathcal{F}} F(x)$$

are weakly increasing in x .

Interpretation. The seller cannot sell more cash flow than the asset produces, and both sold and retained claims must look like sensible asset-backed securities. These conditions are the multi-security version of limited liability and monotonicity.

2.5 Non-exclusive contracting game

The game has three stages:

- **Stage 1.** Buyers simultaneously post menus of contracts, where a contract is a security–price pair (F, p) .
- **Stage 2.** After observing all menus, buyers may withdraw their own menu at a small cost $\kappa > 0$.
- **Stage 3.** The seller observes the active menus and can accept at most one contract from each menu, subject to the accepted collection being feasible.

The seller’s date-1/date-2 payoff from accepting a set of contracts $\mathcal{C}$ is

$$u_\theta(\mathcal{C}) = \sum_{(F,p) \in \mathcal{C}} p + \delta E_\theta \left[X - \sum_{(F,p) \in \mathcal{C}} F(X) \right].$$

Interpretation. The seller can trade bilaterally with many buyers. This one feature—not being able to commit to exclusivity—is what overturns the standard signalling result.

2.6 Why the withdrawal stage matters

- If buyers could not withdraw menus after seeing the market, loss-making offers could survive and equilibrium may fail to exist.
- Allowing withdrawal at a tiny cost plays the same role as in Wilson–Miyazaki style refinements in insurance markets.

- In this paper it is essential for supporting cross-subsidizing equilibria and latent contracts.

Interpretation. The withdrawal stage is not cosmetic. It is the equilibrium device that lets the authors support semi-pooling allocations and rule out cream-skimming deviations.

2.7 Benchmark objects used throughout the paper

The paper compares three environments:

- **Full information benchmark:** buyers observe θ .
- **Exclusive markets benchmark:** the seller can trade with at most one buyer.
- **Non-exclusive markets:** the seller can trade with multiple buyers.

The central equilibrium objects are:

- the cash flows sold by each type,
- the prices at which those claims are sold,
- the amount of retention by the high type,
- market liquidity (how much of the asset gets sold),
- and welfare/origination incentives.

Interpretation. Everything in the paper is a comparison of these objects across market structures, holding the underlying asset environment fixed.

3 Models and empirical specifications (all equations + interpretation)

3.1 Full-information benchmark

If buyers observe the seller's true asset quality before posting menus, the equilibrium is first best.

Proposition 1.

- The seller transfers all asset cash flows.
- The claim is priced at full-information expected value, $E_\theta[X]$.

Interpretation. Non-exclusivity by itself does not cause distortions. The interesting effects come from the interaction of non-exclusivity with adverse selection.

3.2 Exclusive markets benchmark: program $P1$

When the seller can accept at most one menu, the equilibrium allocations solve an optimization program that maximizes the high type's payoff subject to incentive-compatibility and participation

constraints:

$$\max_{(F_H, p_H), (F_L, p_L)} p_H + \delta E_H[X - F_H(X)]$$

subject to

$$\begin{aligned} p_L + \delta E_L[X - F_L(X)] &\geq p_H + \delta E_L[X - F_H(X)], \\ p_H + \delta E_H[X - F_H(X)] &\geq p_L + \delta E_H[X - F_L(X)], \\ \mu_0(E_H[F_H(X)] - p_H) + (1 - \mu_0)(E_L[F_L(X)] - p_L) &\geq 0, \\ p_L &\geq E_L[F_L(X)]. \end{aligned}$$

Interpretation of the constraints.

- The first two are the low-type and high-type IC constraints.
- The third is the buyers' break-even condition.
- The fourth says the low type must get at least her full-information outside option.

The crucial feature of the exclusive benchmark is that if a type accepts a contract, any unsold cash flows are *retained*. This is what makes retention a viable screening device.

3.3 Exclusive-markets solution: separation or cross-subsidization

Lemma 1. The unique solution to $P1$ has two regimes.

- **Low prior on high quality** ($\mu_0 \leq \hat{\mu}$): equilibrium is separating.
 - The high type issues debt:

$$F_H(X) = \min\{d^S, X\}, \quad p_H = E_H[\min\{d^S, X\}].$$

- The low type sells the full claim:

$$F_L(X) = X, \quad p_L = E_L[X].$$

- **High prior on high quality** ($\mu_0 > \hat{\mu}$): equilibrium includes cross-subsidization.
 - The high type still issues debt, but with a larger face value $d^C > d^S$.
 - The high-type debt is priced below high valuation.
 - The low type still sells the full claim, but now at a price above low valuation.

Why debt? Debt minimizes the valuation wedge between types among feasible securities, so it is the cheapest way for the high type to separate or limit the subsidy she gives the low type.

Proposition 2. In exclusive markets, equilibrium exists, is unique, and coincides with the solution to $P1$.

Interpretation. This is the familiar exclusive-contracting logic: the high type can be screened by forcing her to retain the tail of the asset. At high priors, however, even exclusive competition can generate some cross-subsidization.

3.4 Non-exclusive markets: why the IC constraint changes

Once the seller can trade with multiple buyers, the relevant low-type IC constraint is no longer the exclusive one. If the low type mimics the high type's senior security, she can also sell the residual cash flow to another buyer.

The relevant optimization problem becomes

$$\max_{(F_H, p_H), (F_L, p_L)} p_H + \delta E_H[X - F_H(X)]$$

subject to

$$\begin{aligned} p_L + E_L[X - F_L(X)] &\geq p_H + E_L[X - F_H(X)], \\ p_H + \delta E_H[X - F_H(X)] &\geq p_L + \delta E_H[X - F_L(X)], \\ \mu_0(E_H[F_H(X)] - p_H) + (1 - \mu_0)(E_L[F_L(X)] - p_L) &\geq 0, \\ p_L &\geq E_L[F_L(X)]. \end{aligned}$$

Key difference. The low type now evaluates the unsold part of the mimicked contract at *market value*, not at retention value. This is the whole point: under non-exclusivity, retention is not enforceable.

3.5 Non-exclusive-markets solution: the star-equilibrium

Lemma 2. The unique solution to the non-exclusive program has the form:

$$\begin{aligned} F_H(X) &= \min\{d^*, X\}, & p_H &= E_{\mu_0}[F_H(X)], \\ F_L(X) &= X, & p_L &= E_{\mu_0}[F_H(X)] + E_L[X - F_H(X)]. \end{aligned}$$

This can be implemented as two zero-profit contracts:

- a **senior tranche**

$$F_S(X) = \min\{d^*, X\}, \quad p_S = E_{\mu_0}[F_S(X)],$$

accepted by *both* types;

- a **junior tranche**

$$F_J(X) = \max\{X - d^*, 0\}, \quad p_J = E_L[F_J(X)],$$

accepted only by the low type.

Proposition 3. In non-exclusive markets, there exists an equilibrium in which both types accept the senior contract (F_S, p_S) and, in addition, the low type accepts the junior contract (F_J, p_J) .

Interpretation.

- Separation disappears.
- The high type pools with the low type on the senior debt contract.
- Only the low type unloads the junior tranche.
- Standard debt remains optimal, but now as the best way to minimize mispricing in a semi-pooling equilibrium.

3.6 Latent contracts and why equilibrium exists

The star-equilibrium is sustained by *latent contracts*: offers that are posted but not accepted on the equilibrium path.

A candidate equilibrium looks like this:

- two buyers post the senior contract at average valuation;
- two buyers post menus that will buy *any* feasible claim at low valuation;
- the remaining buyers post the trivial contract.

Why this deters deviations.

- A deviation that tries to attract only the low type must pay above low valuation and loses money.
- A deviation that tries to attract only the high type also attracts the low type, because the low type can pair the deviating contract with a latent contract that buys the residual piece.
- A deviation that attracts both types is unprofitable because the senior–junior allocation already solves the relevant optimization problem.

Interpretation. This is the deep logic of the paper. The reason separation fails is not merely that the low type wants to mimic—it is that the market structure lets her complete the mimicry by selling the piece she was supposed to retain.

3.7 Uniqueness and off-equilibrium refinement

Proposition 4. The star-equilibrium is unique if either:

- μ_0 is high enough that the high type sells the full claim in the star-equilibrium, or
- we impose the refinement that any feasible off-equilibrium security must be priced weakly above low valuation.

Interpretation. The refinement effectively says that off-equilibrium prices should still be competitive. Under that discipline, there is no room to support alternative off-equilibrium pricing patterns that generate other equilibria.

The paper also notes that when these conditions fail, other equilibria may exist. They still look like the star-equilibrium, but with extra cash flows issued by the high type and even more cross-subsidization toward the low type.

3.8 Welfare when asset quality is exogenous

Ex-ante welfare is the seller’s expected payoff:

$$W(\mu_0) = \mu_0 u_H(\mu_0) + (1 - \mu_0) u_L(\mu_0).$$

The paper shows a useful decomposition:

$$u_L(\mu_0) = E_L[X] + \Delta(\mu_0),$$

$$u_H(\mu_0) = E_H[X] - (1 - \delta)E_H[X - F_H(X)] - \frac{1 - \mu_0}{\mu_0}\Delta(\mu_0),$$

where $\Delta(\mu_0)$ is the mispricing transfer from the high type to the low type.

Combining the two,

$$W(\mu_0) = W^{FB}(\mu_0) - \mu_0(1 - \delta)E_H[X - F_H(X)].$$

Interpretation.

- Mispricing is a redistribution between types.
- Ex-ante efficiency is distorted only by *retention*, that is, by unrealized gains from trade.
- Hence, when average quality is exogenous, the more liquid market is the more efficient market.

Proposition 5. There exist thresholds $\mu < \bar{\mu}$ such that:

- for low average quality, exclusive markets dominate;
- for intermediate average quality, non-exclusive markets dominate;
- for sufficiently high average quality, the two market structures coincide.

Interpretation. Non-exclusivity is not unambiguously good or bad for welfare once quality is fixed. Its effect runs through liquidity.

3.9 Endogenous quality and origination incentives

The paper then lets the seller choose costly effort q that makes the asset high quality with probability q , at cost $C(q)$.

Ex-ante welfare becomes

$$W(\mu_0) = \max_q \{q u_H(\mu_0) + (1 - q) u_L(\mu_0) - C(q)\},$$

and in equilibrium beliefs satisfy

$$\mu_0 = q^*.$$

The optimal effort condition is

$$C'(q^*) = E_H[X] - E_L[X] - (1 - \delta)E_H[X - F_H(X)] - \frac{\Delta(\mu_0)}{\mu_0}.$$

Interpretation.

- The incentive to originate a high-quality asset depends on the payoff gap between the high and low types.
- Retention lowers this gap because high types cannot monetize all cash flows.
- Mispricing also lowers this gap because the high type subsidizes the low type.

Proposition 6.

$$0 < \mu_0^{NE} < \mu_0^E < \mu_0^{FB} < 1.$$

So average originated asset quality is always lower in non-exclusive markets than in exclusive markets, and lower in exclusive markets than under full information.

Economic message. Even if non-exclusive markets sometimes create more liquidity, they *always* worsen origination incentives because they always induce more mispricing.

3.10 Positive predictions for real-world securitization markets

The paper closes the model section with four empirical predictions.

- **Prediction 1.** The harder exclusivity is to enforce, the more likely cash flows are to be split into separate tranches and sold separately.
- **Prediction 2.** In non-exclusive markets, the amount of retained cash flow should *not* predict senior-tranche pricing.
- **Prediction 3.** In non-exclusive markets, the amount of retained cash flow *should* predict senior-tranche quality.
- **Prediction 4.** As exclusivity becomes harder to enforce, originated asset quality should decline.

Interpretation. These are the paper's main bridges from theory to the CMBS/CDO evidence: pooling on senior claims breaks the link between retention and prices, but not between retention and eventual performance.

4 Identification and instruments (what makes the estimates credible)

4.1 What makes the comparison credible in a theory paper

- There is no instrument or empirical identification design here.
- Credibility comes from a tightly controlled comparison: start from the same adverse-selection environment, change only whether the seller can contract exclusively, and solve the equilibrium in each market structure.
- The paper repeatedly compares three environments—full information, exclusive markets, and non-exclusive markets—so the reader can see exactly which results are due to which friction.

4.2 Why reverse causality is not the issue

- In an empirical paper, one would worry that issuance structure and asset quality jointly determine each other.
- Here, the baseline results are equilibrium statements inside a fully specified model.
- The relevant question is not reverse causality but whether the model's constraints correctly capture what non-exclusivity changes.

Interpretation. The paper’s credibility stands or falls on whether the modified IC constraint is the right one once multiple buyers can buy different pieces of the same asset.

4.3 Why separating contracts fail under non-exclusivity

The core identification-style argument is conceptual:

- In exclusive markets, the high type can separate by retaining a costly tail.
- In non-exclusive markets, another buyer can offer to buy exactly that tail.
- Therefore, any contract that was supposed to separate the high type can be imitated by the low type at lower cost.

Interpretation. This is the key “why we should believe the result” step. The paper is not merely asserting semi-pooling; it derives it from the fact that retention cannot be contracted upon when trades are bilateral and non-exclusive.

4.4 Why debt remains optimal

- The paper does *not* overturn the classic debt-optimality result.
- Instead, it shows that debt remains the best feasible instrument in both exclusive and non-exclusive markets because it minimizes the type-valuation wedge.
- What changes is not the preferred shape of the security, but whether that security can separate types.

Interpretation. The paper’s message is subtle: non-exclusivity kills separation, not debt itself.

4.5 Role of latent contracts and the withdrawal stage

- Latent contracts are essential because they let the low type complete a deviation by selling any residual claim at low valuation.
- Menu withdrawal is essential because buyers who would otherwise lose money after a deviation can pull back their menus.
- Together, these two ingredients make cream-skimming deviations unprofitable and support equilibrium existence.

Interpretation. Without these pieces, one might wrongly conclude that non-exclusive markets either have no equilibrium or revert to a simpler Akerlof-style pooling outcome.

4.6 What is assumed and what is derived

Assumed:

- two seller types,

- competitive, risk-neutral buyers,
- seller impatience,
- MLRP,
- feasible securities satisfying capacity and monotonicity,
- bilateral non-exclusive menus,
- small withdrawal cost.

Derived:

- full-information first best,
- unique exclusive benchmark equilibrium,
- semi-pooling star-equilibrium in non-exclusive markets,
- larger mispricing under non-exclusivity,
- ambiguous liquidity effects,
- lower origination incentives under non-exclusivity.

Interpretation. The strongest results—no separation, more mispricing, lower asset quality—are not put in by assumption; they come out of the equilibrium logic.

4.7 What the paper can and cannot distinguish

- The paper can distinguish clearly between the consequences of exclusivity and non-exclusivity in a stylized screening model.
- It can show when opacity/non-exclusivity raises liquidity and when it lowers welfare.
- It cannot distinguish empirically whether real-world opacity comes from unobservable trades, balance-sheet complexity, legal limits on contracting, or all three simultaneously.
- It also does not build a full quantitative model of the securitization chain or estimate the magnitudes in actual mortgage-backed securities markets.

Interpretation. The paper is strongest as a theory of mechanism, not as a fully estimated quantitative model.

4.8 Relation to the literature

- Relative to classic security-design papers (Leland–Pyle, DeMarzo, Nachman–Noe), the novelty is to remove exclusivity.
- Relative to non-exclusive competition papers such as Attar et al. (2011), the novelty is to let agents design *securities*, not just quantities.
- Relative to non-exclusive insurance models, the paper shows why semi-pooling with a common base contract and type-specific supplemental trade can also arise in asset markets.

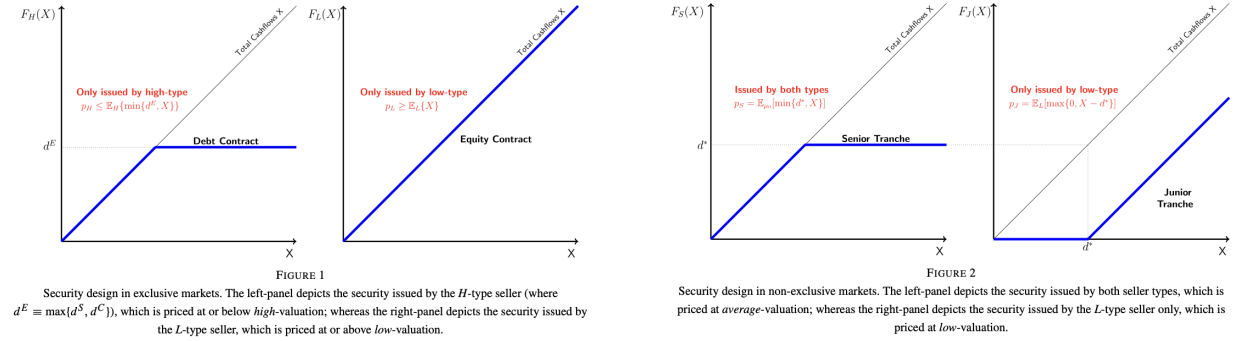
- Relative to transparency/regulation papers, the novelty is to emphasize that transparency matters not only because it reveals signals, but also because it can help enforce exclusivity.

5 Tables and figures

5.1 Figure 1: security design in exclusive markets

Read. The left panel shows the high type issuing a debt contract with face value d^E : the claim rises one-for-one with cash flow until the face value and is flat thereafter. The right panel shows the low type issuing the full claim. When priors are high enough, the same debt shape is kept but priced below high valuation, generating cross-subsidization.

Economic message. Under exclusivity, retention is enforceable, so debt can separate or partially separate types. The high type's ability to keep the tail is exactly what makes this screening logic work.



5.2 Figure 2: security design in non-exclusive markets

Read. The left panel shows a senior tranche sold by *both* types at average valuation. The right panel shows a junior tranche sold only by the low type and priced at low valuation.

Economic message. This is the star-equilibrium. Pooling happens on the senior debt piece, while separation happens only through the low type additionally selling the junior residual cash flow. Non-exclusivity destroys the classic separating contract and replaces it with tranching.

5.3 Figure 3: welfare when average quality is exogenous

Read. The figure plots the debt level in exclusive markets, d^E , and in non-exclusive markets, $d^{NE} = d^*$, against the average quality μ_0 .

Economic message. At low μ_0 , non-exclusive markets force more pooling and less liquidity than exclusive markets. At intermediate μ_0 , they can generate more liquidity than exclusive markets. At high μ_0 , both structures converge because the seller can sell the full claim in either market. This is why welfare with exogenous asset quality can rank the two market structures differently depending on the prior.

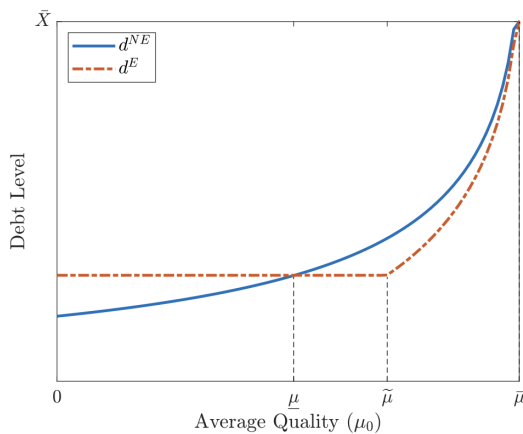


FIGURE 3
Which market structure is more efficient when average quality is exogenous? The figure depicts the debt levels $d^E = \max[d^S, d^C]$ and $d^{NE} = d^*$ that arise with exclusive and non-exclusive markets, respectively, as a function of μ_0 .

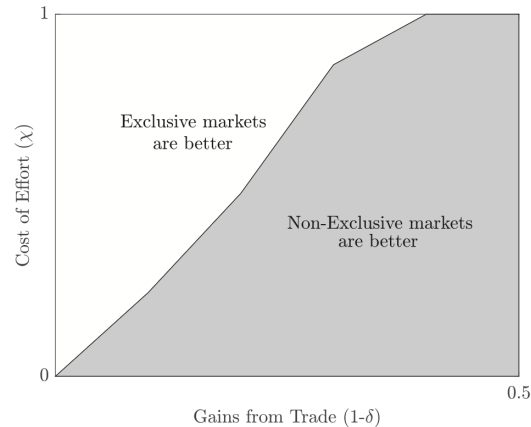


FIGURE 4
Which market structure is more efficient when average quality is endogenous? The unshaded region depicts parameter values for which welfare in the equilibrium with exclusive markets is greater than welfare in the star-equilibrium. The shaded region depicts parameter values for which welfare in the star-equilibrium is greater than welfare in the equilibrium with exclusive markets.

5.4 Figure 4: welfare when average quality is endogenous

Read. The figure splits the $(1 - \delta, \chi)$ parameter space into regions where exclusive markets or non-exclusive markets are more efficient once asset quality is chosen endogenously.

Economic message. Non-exclusive markets only dominate when gains from trade are large enough relative to the cost of effort. Once origination incentives matter, the extra mispricing under non-exclusivity becomes a first-order cost, because it lowers equilibrium asset quality.

5.5 What is missing from Section 5 relative to an empirical paper

- There are no regression tables in this paper.
- The paper is proposition- and figure-driven.
- The core evidence comes from comparative statics and equilibrium constructions, not from estimated coefficients.

6 Short summary

- **Question.** What happens to security design when an informed seller cannot commit to exclusivity?
- **Core mechanism.** Non-exclusivity destroys retention as a credible screening device, because another buyer can always purchase the supposedly retained piece.
- **Exclusive benchmark.** High types use debt and retain the tail; low types sell the full claim.
- **Non-exclusive result.** There is always a semi-pooling star-equilibrium: both types sell the same senior debt at average valuation, and only the low type sells the junior piece.
- **Main welfare result.** With fixed asset quality, non-exclusive markets can be better or worse

depending on liquidity. With endogenous asset quality, they always weaken origination incentives and lower average quality.

- **Empirical relevance.** The model helps explain tranching, the sale of retained pieces, and the weak pricing effect but strong quality effect of retention in mortgage-backed securities markets.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that non-exclusive competition fundamentally changes security design under asymmetric information. In exclusive markets, retention can screen. In non-exclusive markets, it cannot. As a result, equilibrium becomes semi-pooling, debt remains the preferred senior claim, mispricing becomes larger, and origination incentives become weaker.

7.2 Economic intuition

The entire paper turns on one idea: *retention is only informative if it can be enforced*. Exclusive contracts force the high type to keep the tail, which the low type dislikes. Non-exclusive markets undo that logic, because the low type can mimic the high type and then sell the residual piece elsewhere. Once that happens, separating contracts collapse, senior debt becomes a pooling device, and the low type unloads the junior tranche.

7.3 Takeaway

Opacity and non-exclusivity can make markets look more liquid, but they do so by increasing cross-subsidization and compressing incentives to create high-quality assets. The paper's bottom line is therefore not just about tranching. It is about a broader design trade-off in financial markets: the same institutional features that facilitate trade can also undermine screening and weaken origination quality.